Report of Analysis of Replication of Comunale et al. 2021

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Introduction

We attempt to replicate the computational model found in the Comunale et al. 2021 paper, in which a closed-loop lumped parameter model is used to investigate the effect of RV dysfunction on pulmonary and systemic hemodynamics. Specifically, we aim to replicate the healthy state PV loops displayed in the article using the given parameters and hemodynamic model equations.

This circulation model is composed of the ventricular and atria chambers, heart valves, and the pulmonary and systemic vessels. The heart chambers are represented by an elastance model, where a time-varying elastance equation represents myocardial activity. Next, the pulmonary and systemic vessels are simulated through resistance due to blood viscosity and compliance due to vessel elasticity. In addition, the vascular beds were also accounted for through dissipation effects upon the aforementioned resistance. Lastly, the heart valves were represented as a diode associated with resistance, where blood is allowed to flow through only when there is a positive pressure gradient across it. The sets of equations used for this model are shown below in Table 1.

We aim to specifically replicate the healthy left ventricular and right ventricular pressure-volume loops found within the Comunale et al. 2021 paper. To do so, this model was parameterized to represent an average, healthy adult, with the detailed set of parameter values shown below in Table 2. These healthy parameter values are taken from the Supplementary Information of the Comunale et al. 2021 paper. In doing so, I aim to improve my understanding of and skill in cardiopulmonary and hemodynamic computational models in order to work towards a refocused goal of replicating past *in vivo* studies using an *ex vivo* computational model based upon the Kim et al. 2021 circulation model. Matching to the Kim et al. 2021 model, our model uses MatLab R2025a.

Circulation Model Description

Table 1: Terminology

Terminology	Abbreviation
Left Ventricle	LV
Right Ventricle	RV
Left Atria	LA
Right Atria	RA
Pressure-Volume Loops	PV Loops
Pulmonary Arteries	PA
Pulmonary Veins	PV
Systemic Arteries	SA
Systemic Veins	SV
Vascular Bed	VB

Table 2: Circulation Model Equations

Measurement	Equation	Source
Elastance	$E(t) = k(\frac{g1}{1+g1})(\frac{1}{1+g2}) + E_{min}$	Comunale et al. 2021
Contractile Phase	$g_1 = (\frac{t-t_{onset}}{\tau_1})^{m_1}$	Comunale et al. 2021
Relaxation Phase	$g_2 = (\frac{t-t_{onset}}{\tau_1})^{m_2}$	Comunale et al. 2021
k	$k = (E^{max} - E^{min})/max[(\frac{g_1}{1+g_1})(\frac{g_2}{1+g_2})]$	Mynard et al. 2015
Heart Chamber Pressure	$p(t) = E(t)(V(t) - V_{p=0})$	Comunale et al. 2021
Valve & Systemic/Pulmonary Flows	$Q = \frac{P_{up}-P_{down}}{R}$	Garret et al. 2023

Table 3: ODE Equations

Measurement	Equation	Source
Change in Heart Chamber Volume	$\frac{dV}{dt} = Q_{in} - Q_{out}$	Garret et al. 2023
Change in Vessel Pressures	$\frac{dP}{dt} = \frac{Q_{in} - Q_{out}}{C}$	Nigro et al. 2025

Table 4: Heart Value Parameters

Parameter	**LV**	**RV**	**LA**	**RA**	**Units**
E_{max}	2.8	0.45	0.13	0.09	mmHg/mL
E_{min}	0.07	0.035	0.09	0.045	mmHg/mL
$V_{p=0}$	20	30	3	7	mL
τ_1	$0.269T$	$0.269T$	$0.110T$	$0.110T$	sec
τ_2	$0.452T$	$0.452T$	$0.180T$	$0.180T$	sec
m_1	1.32	1.32	1.99	1.99	—
m_2	21.9	21.9	11.2	11.2	—
t_{onset}	0	0	$0.85T$	$0.85T$	sec
R_{valve}	0.01	0.01	0.005	0.005	mmHg * s/mL

Table 5: Model and Initial Parameters

Parameter	Value	Units
$R_{systemic\ arteries}$	0.0448	mmHg * s/mL
$R_{systemic\ vascular\ bed}$	0.824	mmHg * s/mL
$R_{systemic\ veins}$	0.0269	mmHg * s/mL
$R_{pulmonary\ arteries}$	0.003	mmHg * s/mL
$R_{pulmonary\ vascular\ bed}$	0.0552	mmHg * s/mL
$R_{pulmonary\ veins}$	0.0018	mmHg * s/mL
$C_{systemic\ arteries}$	0.983	mL/mmHg
$C_{systemic\ veins}$	29.499	mL/mmHg
$C_{pulmonary\ arteries}$	6.7	mL/mmHg
$C_{pulmonary\ veins}$	15.8	mL/mmHg
$P_{systemic\ arteries}$	100	mmHg
$P_{systemic\ veins}$	4	mmHg
$P_{pulmonary\ arteries}$	15	mmHg
$P_{pulmonary\ veins}$	8	mmHg
$V_{p=0,LV}$	149.6	mL
$V_{p=0,RV}$	189.2	mL
$V_{p=0,LA}$	71	mL
$V_{p=0,RA}$	67	mL

Circulation Model Approach

Our circulation model aims to emulate the Comunale et al. 2021 computational model and create an identical healthy LV and RV PV loop. Parameters from Table 3 and Table 4 were used in order to properly replicate an average, healthy human. We first calculate the k values for each of the 4 heart compartments, which are factors that ensure that the max elastance calculated for each compartment matches to the max elastance found within the given parameters. Then, we calculate the contractile (g1) and relaxation (g2) phases of the cardiac cycle to input into our time-varying elastance model. Through calculating elastance, we can then calculate the pressure of each of the 4 heart chambers.

In order to find volumes of the 4 heart chambers for use in our PV loops, we create a system of ordinary differential equations (ODEs) to solve with the MatLab ode15s command. Primarily, we solve for change in volume and change in systemic and pulmonary vessel pressures, using our previously calculated pressure values. Using our newly calculated chamber volumes, we can then recalculate pressures for the heart chambers to be used for our PV loop figures. Finally, we graph our recalculated pressures and volumes in order to create healthy LV and RV PV loops.

Troubleshooting

One of the greatest challenges in creating this replica model was with the k scaling factor listed among the circulation model equations listed in Table 2. This k scaling factor is placed into our time-varying elastance model to ensure that the max values calculated from this model are equal to the inputted max elastance parameters. Originally, this was represented as a scalar constant, but generated PV loops following this logic were, while of the right shape, showing increased stroke volume and higher volumes than those of the Comunale et al. 2021 model. However, as can be seen from Table 2, this equation was not found in the Comunale et al. 2021 paper, meaning that there was little explanation as to how to implement this equation into the model. For reference, below is the equation for the k scaling factor, from Mynard et al. 2015.

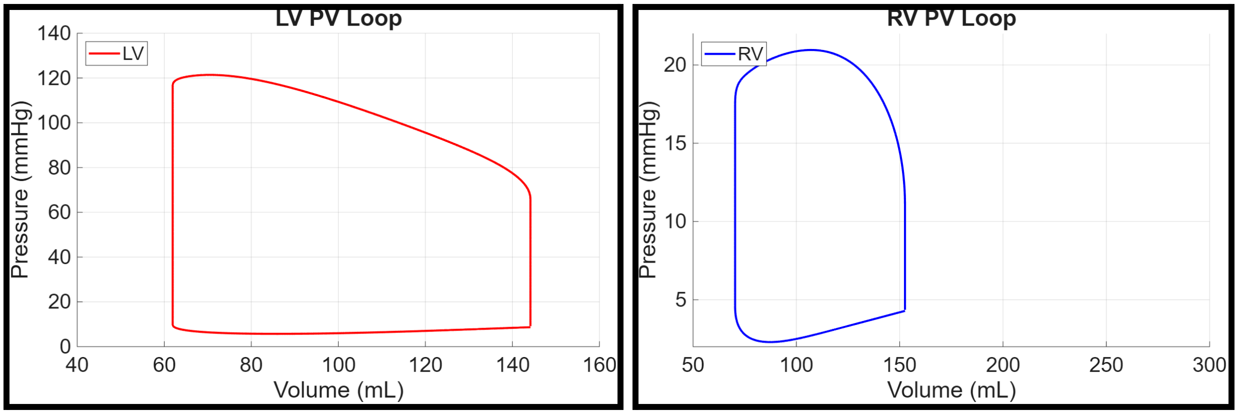
$$k = (E^{max} - E^{min})/max[(\frac{g_1}{1+g_1})(\frac{g_2}{1+g_2})]$$

In integrating this equation into our model, flaws in our model equations were exposed. In order to precalculate our k scaling factor for the rest of our model equations, we had to also precalculate g1 and g2. Due to this, errors within our g1 and g2 model equations were exposed due to now being calculated with a wider range of time steps between the time span of a period. For reference, below is the equation for g1 and g2, from Comunale et al. 2021.

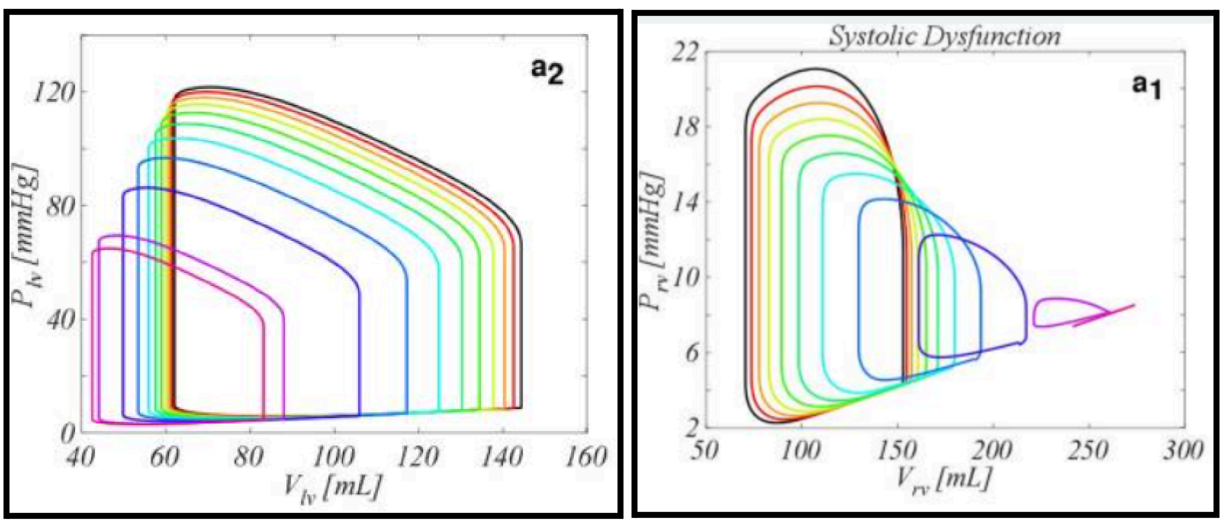
$$g_1 = (\frac{t-t_{onset}}{\tau_1})^{m_1},\ g_2 = (\frac{t-t_{onset}}{\tau_1})^{m_2}$$

When calculated over a larger number of individual time steps, the numerator of this equation, $t - t_{onset}$, would often become a negative number. When calculated to the power of a decimal in m1 or m2, our final g1 and g2 calculations would become complex numbers, which caused errors in our model. In order to resolve this issue, the numerator was individually calculated first and clamped to have a minimum value of 0, ensuring that g1 and g2 are never calculated as complex numbers.

Circulation Model Results



Figures 1: LV (left) and RV (right) healthy PV Loops created from our replica computational model.



Figures 2: LV (left) and RV (right) PV loops from the Comunale et al. 2021 model. Healthy PV loops are drawn with a black line.

Summary of Findings

From comparison between Figures 1 and 2, our goal of replicating the Comunale et al. 2021's circulation model was a success. Our created PV loops appear to match the healthy PV loops from the paper we aimed to match. We can create simulations of RV dysfunction through modifying the parameters we inputted into the model. Specifically, RV systolic dysfunction can be done by reducing max contraction, represented by reducing $E_{max,RV}$, as well as increasing systolic times, which are represented in our equations by $\tau_{1,\,RV}$ *and* $\tau_{2,RV}$. Meanwhile, RV diastolic dysfunction can be simulated by increasing ventricular wall stiffness, which can be represented in our model by increasing $E_{min,RV}$ and decreasing $m_{2,\,RV}$.

From this project, I've learned essential skills in research and gathering of data, understanding of physiological equations, as well as in hemodynamic computational modelling and the difficult troubleshooting that comes with it. The circulation model MatLab code can be found on github, at [LINK].

The Comunale et al. 2021 paper has already explored applying this circulation model towards investigating the effects of varying severities of RV dysfunction. However, the paper also discusses possible improvements upon the model that could be worked on. For example, a limitation listed by the paper is that the simple linear relationship representing cardiac valve flow has been found to lead towards aortic and mitral valve flows having higher peaks than in physiological studies. As a future step, we could begin working upon eliminating this limitation, following along with the paper's belief that this issue can be resolved through including more complex flow and pressure gradient relationships for cardiac valve flow.

In addition, with our main research investigating the effects of ventricular interdependence, this circulation model can aid in our investigations as a quick and easy simulation for testing smaller variable changes.

Sources

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